Language Selects, Geometry Computes: Systematic Generalization in Language-Conditioned Molecular Observables

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Abstract

A chemist asks a model "what is the force on the carbon bonded to two hydrogens, minus the force on the oxygen?" and expects a vector that rotates with the molecule. We study which *interface* between language and an O(3)-equivariant molecular model produces such answers and, above all, which one keeps working when the question is worded in a new way, asks about a new scaffold, or combines an entity with an operator never seen together. We introduce TensorQA-v1, a benchmark of force, difference and projection questions with leak-aware splits (canonical-graph de-duplication, scaffold hold-out, held-out wording families written by an LLM and audited by a second model), counterfactual consistency checks, and exact tests under rotations *and* reflections; and TensorQA-pol, language-chosen contractions of coupled-cluster polarizability tensors ($\ell = 0 \oplus 2$). We compare end-to-end text-to-coefficient readouts, a learned parser with an exact executor, and a hand-written rule parser with an exact executor, and decompose every error into a *physical-quantity* part and a *program* part. [TODO: one-sentence headline result once the runs finish.] We also report two negative findings that change how such models should be motivated: with perfect per-atom forces a single convex combination already reaches cosine 0.97–0.99 on our compositional targets, so the gap is not one of expressivity; and the common practice of feeding signed odd-parity scalars to a scalar network silently breaks reflection equivariance of vector outputs.

1 Introduction

Equivariant networks give exact transformation laws for geometric outputs, and language models give flexible access to what a user wants. Putting them together raises a question that is not about symmetry at all: when the question is stated in words, who decides *which* physical observable, on *which* atoms, combined *how*? Three answers are in use. The language may produce a program that a classical parser or an exact executor runs on equivariant features; the text may be mapped directly to the coefficients of a readout over equivariant vectors; or a large language model may take the geometry as invariant context and let an equivariant adapter produce directions [Li et al., 2025]. Related mechanisms exist: invariant queries that attend to equivariant values [Chen et al., 2024], language-conditioned SE(2) and SE(3) policies in robotics [Jia et al., 2024, Zhu et al., 2025], and the observation that equivariant functions can be written with scalar coefficients on geometric generators [Villar et al., 2021]. None of them asks the question we ask: *which interface generalizes systematically in a scientific question-answering setting, and where does its error come from?*

Contributions. (1) TensorQA-v1 and TensorQA-pol, with leak audits of the earlier split, held-out wording families, scaffold and combination splits, counterfactual identities and reflection tests (Section 3). (2) A controlled comparison of four interfaces on the same equivariant backbone and the same label budget, with a 2×2 error decomposition (Section 6). (3) Two corrections to common beliefs, with proofs and measurements (Section 5). (4) Code, data generators, audited templates and checkpoints.

2 Related work

Equivariant readouts and cross-attention. GeoMFormer [Chen et al., 2024] exchanges information between an invariant and an equivariant stream with cross-attention; our attention heads are of the same family, with the query coming from text. NequIP-style message passing [Batzner et al., 2022, Geiger and Smidt, 2022] supplies our encoder. **Language and equivariance.** EquiLLM [Li et al., 2025] keeps all directional computation outside the language model; GEM [Jia et al., 2024] and EquAct [Zhu et al., 2025] condition equivariant policies on text in the plane and in SE(3). We differ in the object studied (scientific observables with typed outputs, atoms referred to in words) and in the evaluation (out-of-distribution wording, entity, operator and scaffold). **Scalars suffice.** Villar et al. [2021] show that equivariant functions can be expressed through scalar contractions; hence "a text model only sees scalars" is not a limitation by itself, and we do not use it as motivation. **Molecular tensors.** Polarizability tensors from coupled-cluster theory exist for QM7b [Yang et al., 2019, Wilkins et al., 2019]; we turn them into language-addressed contractions.

3 Benchmark

TensorQA-v1. The molecules are the 20k QM9 [Ramakrishnan et al., 2014] geometries with GFN2-xTB [Bannwarth et al., 2019] forces; the labels therefore test the *interface*, not electronic-structure accuracy ($0.0064\,\mathrm{Eh/bohr} \approx 0.33\,\mathrm{eV/Å}$). A question names atoms by element, ring membership and neighbour counts, and asks for one of three vector answers: `force` F_A; `diff` $F_A - F_B$; `proj` $(F_A u)u$ with $u = (r_B - r_A)/\|r_B - r_A\|$. `proj` is the *signed axis projection*: it points away from B when $F_A \cdot u < 0$.

Splits and what they test. The random molecule split of an earlier version shared 4 canonical heavy-atom graphs between splits and 86.1% of its test molecules had a Bemis–Murcko scaffold seen in training (1710/1986), so it could not support an out-of-distribution claim. v1 therefore (i) groups molecules by canonical graph, (ii) holds out whole scaffold groups (`scaf`), (iii) re-words test questions in two *held-out style families* (`test_heldout`, `scaf_heldout`), (iv) holds out the combinations (`proj`, A is O) and (`diff`, A is N) from all training data (`comb`), and (v) adds counterfactual groups (`cf`) on which a correct answerer must satisfy $D(A, B) = -D(B, A)$ and $D(A, C) = D(A, B) + D(B, C)$.

Wording. Six style families (textbook, terse, technical, indirect, chatty, inverted) of templates with atom placeholders were written by `claude-haiku-5-5`, filtered by a second call, and audited by `claude-sonnet-5-5`; 66 templates were removed (e.g. "bond axis" for two atoms that need not be bonded, or an unspecified order of subtraction). This is not a human audit; a 198-question sheet for one is released. Chatty and inverted appear only in held-out splits.

TensorQA-pol. 7211 QM7b molecules with LR-CCSD/d-aug-cc-pVDZ static polarizability tensors [Yang et al., 2019]. Questions ask for α_{iso} ($\ell = 0$), the traceless part ($\ell = 2$, five coefficients), $u^\top \alpha u$ along the direction from atom A to atom B, and the induced dipole αu ($\ell = 1$). The B3LYP tensor of the same molecule is reported as a classical reference.

Symmetry tests. Every model is tested on random rotations *and* reflections (improper orthogonal maps), with outputs transformed by the matching representation (R for vectors, the Wigner matrix for $\ell = 2$, trivially for scalars).

4 Interfaces

All models share the encoder: a three-layer NequIP-style network ($\ell \leq 2$) on atomic numbers and positions, giving irreps features x_i per atom, and a frozen text encoder (MatSciBERT [Gupta et al., 2022]) run online. Write $V_i = W x_i \in \mathbb{R}^3$ for the per-atom vector readout.

- **Option A.** One attention head: $a = \mathrm{softmax}(q(\mathrm{text}) \cdot k(I(X)))$, $y = \sum_i a_i V_i$ (+ a gated sum).

- **B/C-lite.** Two pointer heads give $F_k = \sum_i a_{ki} V_i$ and positions P_k; with $u = (P_2 - P_1)/\|P_2 - P_1\|$, $y = c_1 F_1 + c_2 F_2 + c_3 (F_1 \cdot u) u + c_4 (F_2 \cdot u) u$ and $c = c(\text{text})$ invariant.
- **Learned parser + exact executor.** Heads predict the task and two atom pointers (supervised by the program), and the executor applies F_1, $F_1 - F_2$ or $(F_1 \cdot u)u$. Hard decisions at test time.
- **Rule parser + exact executor.** A hand-written parser maps the question to (task, A, B). We wrote its rules in two rounds, reading failures on the *train* split only; its accuracy on held-out families is reported.
- **Oracle program** (diagnostic): the true program with the same executor.

The O(3) variants use $|x|$ for odd scalar channels in $I(X)$ (Section 5); the "so3" variant keeps their sign.

5 Theory: three small statements

Proposition 1 (closure)**.** *Let $V_i(X)$ be* O(3)*-equivariant vectors, $I(X)$ any* O(3)*-invariant features of X, q any text features and $a = \text{softmax}(f(q, I(X)))$. Then $y = \sum_i a_i V_i$ is* O(3)*-equivariant, as is any fixed polynomial in dot products and sums of such y's (difference, axis projection $(F \cdot u)u$, contraction of a tensor with u).*

Proposition 2 (parity leak)**.** *If $I(X)$ contains an odd scalar channel s with its sign ($s(PX) = -s(X)$ for a reflection P), then in general $y(PX) \neq Py(X)$ although $y(RX) = Ry(X)$ for all rotations R. Replacing s by $|s|$ restores exact* O(3)*-equivariance.*

Sketch. The first statement is composition of equivariant maps. For the second take $f = \tanh(s)$ and a over two atoms related by P; the weights differ between X and PX so $y(PX)$ is not the reflection of $y(X)$. □

Proposition 3 (ceiling of one pointer)**.** *For any weights $a_i \geq 0$, $\sum_i a_i = 1$: $y \in \text{conv}\{V_i\}$. With the exact forces $V_i = F_i$ the best achievable answer to a question with target t is the Euclidean projection of t onto* $\text{conv}\{F_i\}$.

Prop. 3 is a computable ceiling (Section 6); it turns out to be high, so the difference between interfaces is about what is *learnable* from text and invariants, not what is *representable*.

Proposition 4 (error decomposition)**.** *For an executor E_p that is Lipschitz in the per-atom field and programs p (true), $\hat{p}$ (predicted), fields V (true), $\hat{V}$: $\|E_{\hat{p}}(\hat{V}) - E_p(V)\| \leq \|E_{\hat{p}}(\hat{V}) - E_{\hat{p}}(V)\| + \|E_{\hat{p}}(V) - E_p(V)\|$.*

The first term is the *quantity error* (right program, imperfect field), the second the *program error* (perfect field, wrong program); the 2×2 table reports both.

6 Experiments

[TODO: filled from `results_v1.md` and `results_pol.md` after the runs finish: interface table; 2x2; wording families; reflection and parity; counterfactual; budget curves; wording coverage; polarizability.]

7 Limitations

The wording is LLM-generated and only audited by another LLM; the sheet for a human audit is released. Labels are GFN2-xTB forces (an interface test, not DFT accuracy) and CCSD polarizabilities for small molecules (up to 7 heavy atoms). Atoms are referred to by unique descriptions, so ambiguous atoms are never asked about. The rule parser was developed by us and could be improved; we report its accuracy and the exact rounds of development. Program depth beyond two operators, new operators at test time, crystals and generation are not studied.

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